SECTION A: 40 MARKS

Answer **all** *the questions in this section.*
Questions **1** *to* **20** *carry* **two** *marks each.*

1. Work out:
$$\begin{array}{r} 385 \\ -264 \\ \hline \\ \hline \end{array}$$

2. State the digit in the hundreds place value in the number 60,791.

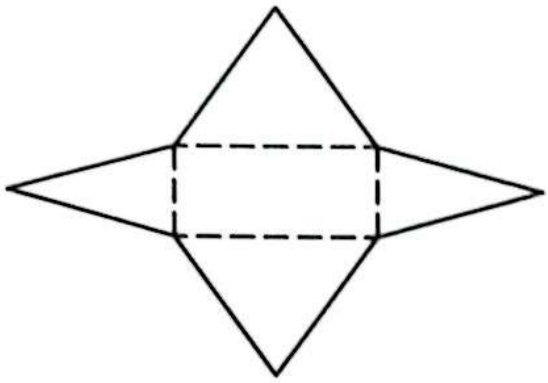
3. Find the next number in the sequence:

5, 9, 18, 34, ____

4. Given that A = {m, n, g, z}, find the number of subsets in set A.

5. Work out: $3\frac{1}{5} + \frac{1}{4}$

6. Name the solid formed from the net below.

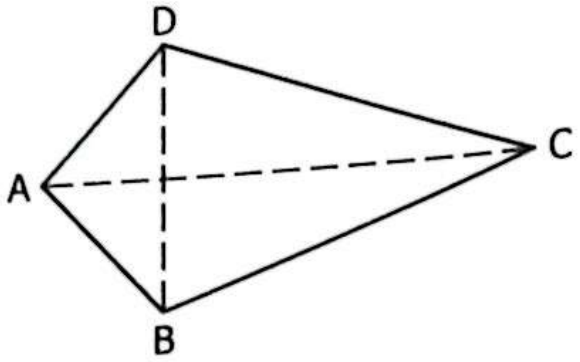


7. Solve: $18 + 2x = 5x$

8. Represent $^{-}4 + {}^{+}6$ on a number line.

9. Write the number whose prime factorisation is $\{2_1, 2_2, 7_1\}$.

10. The figure ABCD below is a kite. The diagonals AC = 12 cm and BD = 7 cm.



Calculate the area of the kite.



11. In a test, two pupils scored 10 marks each, four pupils scored 7 marks each and one pupil scored 12 marks. Find their median mark.

12. Okumu bought two similar balls at sh 35,000. He later sold the balls and made a profit of sh 3,000. Given that he sold each of the balls at the same price, find the selling price of each ball.

13. Complete the base five multiplication table below. (Show your working)

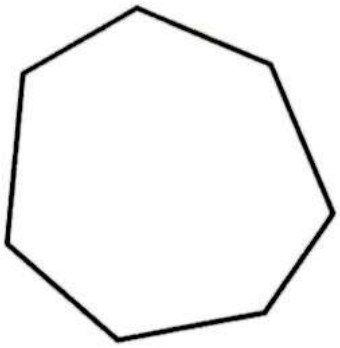
×	2	3	4
1	2	3	4
3	.....	14	.....

14. Babu and his sister contributed money for their mother's birthday in the ratio 3:4 respectively. Babu contributed sh 150,000. Calculate the amount of money the sister contributed.

15. Change the time 1:35 a.m. to 24 - hour clock.

Turn Over

16. Calculate the interior angle sum of the polygon below.

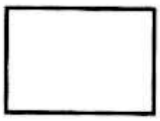


17. Rogers is 5 years younger than Aida. Write an algebraic expression for the sum of the ages of Rogers and Aida.

18. Maneno had some mangoes to sell. When she put the mangoes in heaps of 5, she remained with 4 mangoes. When she again put them in heaps of 6, only 5 mangoes remained. Find the least number of mangoes Maneno could have had.

19. A patient received 75 ml of medicine for cough. The patient was told to take a dose of 5 ml of the medicine three times each day. Find the number of days the medicine lasted.

20. Smith had 250 British pounds. He exchanged all the pounds to Uganda shillings and got sh 1,200,000. Calculate the rate of exchange for the British pounds in Uganda shillings.



SECTION B: 60 MARKS

Answer **all** *the questions in this section.*

Marks for each question are indicated in the brackets.

21. A pupil formed three-digit numbers using the digits 2, 1 and 6 without repeating a digit in each number formed.

Write down the;

(a) three-digit odd numbers formed. (*03 marks*)

(b) three-digit numbers divisible by 4. (*02 marks*)

22. (a) Round off 52.479 to the nearest tenth. (*01 mark*)

(b) Work out: $\frac{0.69 \times 1.2}{0.43 - 0.2}$ (*03 marks*)

23. There are 60 farmers in a village. 32 of them use oxen (X) to prepare land while 28 use tractors (T) to prepare their land. 15 farmers use both oxen and tractors. The number of farmers who use neither oxen nor tractors is 2 less than those that use oxen only.

(a) Draw a Venn diagram to represent the given information. (*04 marks*)

(b) Find the probability that a farmer selected at random from the village does not use oxen to prepare land. (*02 marks*)

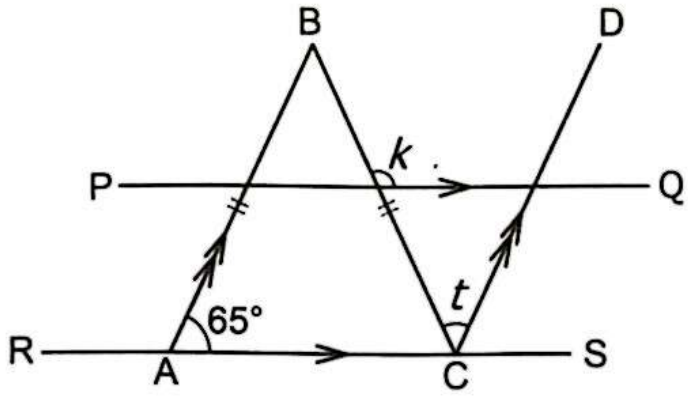
24. Agnes makes juice from passion fruit. She mixes the passion juice with some water before serving it. In her mixture, the passion juice is 40% more than the water she adds.

(a) Calculate the percentage of water she adds. *(02 marks)*

(b) Given that Agnes has 14 litres of passion juice to be mixed with water, find in litres, the amount of mixture she will prepare. *(03 marks)*

25. The mass of a box containing 12 similar textbooks is 7.2 kg. When all the textbooks were removed, the mass of the empty box is 240 g. Calculate the mass of each textbook. *(04 marks)*

26. In the diagram below, ABC is an isosceles triangle. Line PQ is parallel to line RS and line AB is parallel to line CD. Angle BAC = 65°.
Study the diagram and use it to answer the questions that follow.



Calculate the size of;

(a) angle t. *(02 marks)*

(b) angle k. *(02 marks)*

27. Musa bought the following items to make mandazi:

2 litres of cooking oil at sh 14,000.
500 grams of sugar at sh 3,000.
2 packets of wheat flour at sh 18,000.
A packet of baking powder at sh 1,200.

Musa made 90 mandazi from the items. He sold each mandazi at sh 500. Calculate the profit Musa made. *(05 marks)*

28. A father paid school fees for his three children; Moses, Sarah and Isaac. The school fees for Moses was sh 30,000 less than that of Sarah while that of Isaac was twice as much as that of Sarah. The school fees for Moses was sh 140,000 less than that of Isaac.

Calculate the school fees paid for each child. *(05 marks)*

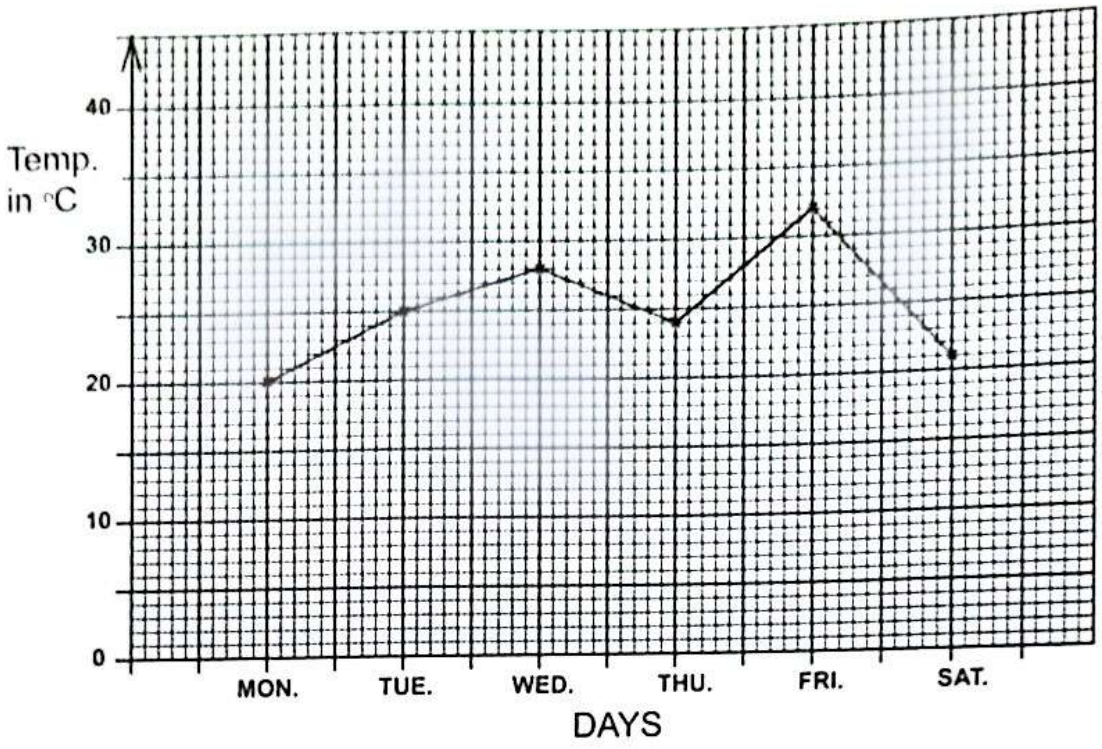
29. Town **K** and town **L** are 200 km apart. Abdul left town **K** at 8:00 a.m. and drove at a steady speed of 60 km/h to town **L**. Abbey left town **L** at the same time driving to town **K** along the same road. After covering 50 km in one hour, Abbey got a flat tyre which took him 30 minutes to repair. He then continued for $2\frac{1}{2}$ hours to town **K**. The two drivers met at 10:00 a.m.

(a) Find the time at which Abbey reached town **K**. (*02 marks*)

(b) Calculate how far Abbey was from town **L** when he met Abdul. (*03 marks*)

30. A trader sells honey using cylindrical containers of diameter 10 cm and height 7 cm. The trader sells each full container of honey at sh 7,500. On one market day, the trader earned sh 300,000 from the sale of honey. Calculate the number of litres of honey the trader sold that day. (use $\pi = \frac{22}{7}$) (*05 marks*)

31. The graph below shows the temperatures of a town recorded in six days. Study and use it to answer the questions that follow.



(a) On which day was the temperature of the town above 30 °C? (*01 mark*)

(b) Find the range of the temperatures recorded. (*02 marks*)

(c) Calculate the average temperature of the town recorded in the six days. (*03 marks*)

32. An aeroplane flew from airport **K** to airport **M** which is 1,800 km away in Northwest direction. The aeroplane then left airport **M** for airport **T**, a distance of 2,400 km on a bearing of 235°.

(a) Using a scale of 1 cm to represent 300 km, draw an accurate diagram to show the route of the aeroplane. (*05 marks*)

(b) Find the bearing of airport **T** from airport **K**. (*01 mark*)

END